Wong Jun Xiong

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linkedin.com/in/wong-jun-xiong-bbbb662a1 | github.com/Junxiong10 | Wong Jun Xiong Peronal Portfolio

Education

Asia Pacific University (APU) **June 2022 – June 2025**
Bachelor of Computer Science (Hons) (Artificial Intelligence) *Bukit Jalil, Kuala Lumpur*

Work Experience

Beyondsoft Malaysia **October 2025 – Present**
AI Engineer *KL Eco City, Kuala Lumpur*

- Conducted testing and evaluation of Microsoft Fabric Data Agent, validating its capability in handling Power BI semantic models, data queries, and automated insights generation, while designing and executing test scenarios using prompt engineering.
- Optimized LLM-driven Fabric Data Agent performance through prompt engineering and agent instruction design, improving response accuracy, grounding quality, and contextual relevance against Power BI semantic model datasets.
- Leveraged prompt engineering with GitHub Copilot Enterprise and Model Context Protocol (MCP) servers to enable LLM-driven automation and structured task execution across Microsoft ecosystem services.
- Leveraged GitHub Copilot Enterprise and MCP servers to build scalable data engineering pipelines and design modular AI agents with reusable SKILLs for task-specific automation.

YToday Sdn Bhd **April 2024 – August 2024**
IT & Project Management Internship *Seri Kembangan, Selangor*

- Collaborated with stakeholders to support IT enhancements through requirement gathering, UI/UX improvements, user role design, and UAT.
- Contributed to an NLP-based chatbot to enhance customer interaction and efficiency.
- Applied problem-solving skills to address project challenges and support the smooth execution of multiple projects, ensuring timely completion, proper documentation, and progress tracking.

Projects

Agentic RAG - AI/ML Knowledge Assistant | *Python · Streamlit · LangChain · ChromaDB* **May 2026**

- Built an agentic RAG system using LangGraph with a self-correcting workflow, including query decomposition, hybrid retrieval, document grading, answer generation, and automatic query rewriting on insufficient results.
- Implemented hybrid search combining dense (ChromaDB) and sparse (BM25) retrieval with Reciprocal Rank Fusion and cross-encoder re-ranking for improved relevance accuracy.
- Deployed a fully local, privacy-preserving architecture using Ollama (Qwen 2.5:7b) with a Streamlit chat UI featuring citation verification, agent trace visualization, and source traceability.
- Link: https://github.com/Junxiong10/Agentic-RAG-AI-ML-Knowledge-Assistant

Product Catalog ReAct Agent | *Python · CLI · LangChain · Hybrid Search · TF-IDF* **May 2026**

- Built a conversational product catalog agent using LangGraph and ReAct (Reasoning + Acting) for multi-step reasoning and autonomous tool selection (search, retrieval, shortlist management).
- Implemented a hybrid search engine combining TF-IDF cosine similarity and keyword matching with weighted scoring and confidence-based result selection.
- Designed a custom LangGraph state reducer for shortlist management that merges by product ID and auto-accumulates quantities at the state layer — keeping tool logic stateless, with hallucination-resilient input parsing for local LLMs and callback-based middleware for real-time token usage monitoring.
- Link: https://github.com/Junxiong10/Product-Catalog-Research-Agent

Restaurant Recommendation System with Sentiment and Emotion Analysis (FYP - 4.0 GPA)
Python · Machine Learning · Web Development · Item-Based CF with Cosine Similarity **April 2025**

- Developed an AI-driven restaurant recommendation system that addresses the lack of sentiment-aware review understanding by analyzing customer reviews using sentiment and emotion classification beyond traditional star ratings.
- Applied machine learning techniques for sentiment classification and emotion analysis on restaurant review data to enable sentiment-aware recommendation intelligence.
- Implemented item-based collaborative filtering using cosine similarity on restaurant sentiment and emotion profiles, leveraging users' bookmarked restaurants to recommend similar dining options.

Technical Skills

Languages: Python, Java, SQL, R, C#, HTML/CSS, JavaScript
Developer Tools: Visual Studio Code, Visual Studio, PyCharm, RStudio, NetBeans, Cursor, GitHub Copilot
Frameworks: Flask, LangChain, FastAPI, Streamlit, React, TailwindCSS
Libraries: pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, NLTK, spaCy, VADER, TextBlob, Hugging Face Transformers, Matplotlib, Seaborn, Plotly